

Bilinear Closure of Generalized Stieltjes Correlations

Gamma-ratio kernels, polylogarithmic jets, exact collision expansions and antiderivatives

Research continuation prepared for Vladimir Reshetnikov's ProveIt project

Mathematical derivations and computational audit with OpenAI assistance

10 October 2026

Abstract

We study products of generalized Stieltjes functions under a circular translation. Their ordinary integrals diverge, but a specified endpoint finite part admits a holomorphic two-parameter generating function. A Gamma-ratio identity proves that every bilinear finite-part correlation is a finite *linear* combination of single Stieltjes values at the translation and its reflection, with coefficients polynomial in positive integer zeta values. The highest Stieltjes order is explicit, and the next order is absent identically. We derive the coincident-endpoint analogue, an exact convergent collision expansion with a canonical logarithmic counterterm, rational-grid and character identities, and explicit antiderivatives. An ordinary, nonregularized log-Gamma autocorrelation also reduces to Hurwitz-zeta derivatives at -1 , equivalently to spectral polylogarithm derivatives at 2.

The proofs use classical Hurwitz Fourier theory and local subtraction; the finite-part closure and its consequences are developed here as a continuation, not asserted to have established literature priority. The package contains exact coefficient identities through total order 8, 272 symbolic checks, and 22 independent numerical diagnostics. A reproducible numerical differentiation canary is recorded separately. Neither floating-point agreement nor the symbolic checks replace the analytic proofs, and no proof-assistant formalization is claimed.

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1 Research position and the main result

The ProveIt manuscript connects polylogarithms, Gamma values, Hurwitz-zeta jets and generalized Stieltjes functions. Its integration chapter already contains the classical single-function ladder that expresses an antiderivative of $\gamma_n(a)$ through $\zeta^{(n+1)}(0, a)$ [4]. A natural next question is whether a product of two Stieltjes functions introduces genuinely bilinear special-function coordinates after integration. For the translated finite parts studied here, the answer is particularly simple: *it does not*.

Here and below, a superscript on ζ differentiates its spectral variable. The functions $\gamma_n(a)$ are fixed by

$$\zeta(1+u, a) = \frac{1}{u} + \sum_{n \geq 0} \frac{(-1)^n}{n!} \gamma_n(a) u^n, \quad \gamma_0(a) = -\psi(a). \quad (1.1)$$

Write $b = 1 - a$ for $0 < a < 1$, and let $I_{mn}(a)$ denote the endpoint finite part specified in Definition 3.1. Our principal result is

$$\begin{aligned} I_{mn}(a) = & \frac{\gamma_{m+n+1}(a)}{m+1} + \frac{\gamma_{m+n+1}(1-a)}{n+1} + e_{mn} \\ & + \sum_{j=0}^{m+n-1} (c_{mn,j} \gamma_j(a) + d_{mn,j} \gamma_j(1-a)), \end{aligned} \quad (1.2)$$

where the sum is empty if $m+n=0$, and all coefficients belong to $\mathbb{Q}[\zeta(2), \dots, \zeta(m+n+2)]$. In particular there is no term of order γ_{m+n} . The statement is an identity for every $m, n \geq 0$, not an extrapolation from the accompanying finite table.

At the first order, the result reads

$$\text{FP} \int_0^1 \psi(x) \psi(x \oplus a) dx = \gamma_1(a) + \gamma_1(1-a) - \frac{\pi^2}{3}. \quad (1.3)$$

Here $x \oplus a$ is addition modulo one, interpreted on the open pieces of the interval. The symbol FP is essential. Removing it makes (1.3) false as a statement about an ordinary integral. An equivalent absolutely convergent form is given in (5.7).

The relevant classical background is the Hurwitz Fourier expansion, its parameter-shift law, and the Lerch identity [1, 2]. Integrals of products of Hurwitz zeta functions have substantial prior literature; in particular Shpot–Chaudhary–Paris treat products of the auxiliary functions $\zeta(s, 1+x)$ and related reflected products [3]. We do not claim the Fourier correlation mechanism itself is new. The specific all-order finite-part closure, its missing-order rule, and the collision-compatible coefficient presentation are the research outputs proved in this article. A comprehensive priority determination would require a wider literature comparison than this delivery provides.

1.1 What is, and is not, settled

This continuation proves exact identities rather than addressing the recent zero-geometry and extremizer problems. It does not prove numerical independence of the resulting constants. Nor does it settle the remaining colored-polylogarithm reduction problems in the manuscript. The inspected current S_4 chapter already contains an exact proof and certificate [5]; it must not be reclassified as an open conjecture on the basis of older research notes. The surviving S_6 reduction problem recorded in the discovery chapter is not resolved here [6].

The repository review was targeted: selected manuscript sections, source metadata, and the incoming directory inventory and intake instructions were inspected. The incoming binary ZIP contents were not audited. No existing repository theorem is declared false in this report. Section 11 identifies normalization and computational hazards that an integration of these new formulas must avoid.

2 Analytic conventions and a translated Hurwitz kernel

For $0 < a < 1$, set $b = 1 - a$ and define

$$C(s, t; a) = \int_0^b \zeta(s, x) \zeta(t, a + x) dx + \int_0^a \zeta(s, b + y) \zeta(t, y) dy. \quad (2.1)$$

This is the circularly translated product, without complex conjugation. All powers of positive real quantities use their real logarithms. At $e^{\pm 2\pi i a}$ we use the standard polylogarithm continuation in its argument, with spectral derivatives taken on that fixed branch. Since $0 < a < 1$, these arguments are never 1.

Lemma 2.1 (Convergence and zero mean). *The integral (2.1) is absolutely convergent and jointly holomorphic for $\operatorname{Re} s < 1$, $\operatorname{Re} t < 1$. In this region,*

$$\int_0^1 \zeta(s, x) dx = 0. \quad (2.2)$$

Proof. Use $\zeta(s, x) = x^{-s} + \zeta(s, 1 + x)$. On the first piece only the factor with parameter x is singular at zero; on the second piece only the factor with parameter y is singular. On compact subsets of the stated spectral region, the other factors and all their spectral derivatives are bounded. The potentially singular terms are dominated by $x^{-\sigma}(1 + |\log x|^k)$ with $\sigma < 1$, or the corresponding expression in y . This proves convergence, holomorphy and differentiation under the integral on compact subsets. Finally, the parameter derivative identity

$$\partial_x \zeta(s - 1, x) = (1 - s) \zeta(s, x) \quad (2.3)$$

and the shift law show that the endpoint values of $\zeta(s - 1, x)$ at zero and one agree when $\operatorname{Re} s < 1$. Integration proves (2.2). $\square$

Theorem 2.2 (Two forms of the translated kernel). *For $0 < a < 1$ and $\operatorname{Re} s, \operatorname{Re} t < 1$, put $z = e^{2\pi i a}$. Then*

$$C(s, t; a) = \Gamma(1 - s) \Gamma(1 - t) (2\pi)^{s+t-2} \left\{ e^{i\pi(s-t)/2} \operatorname{Li}_{2-s-t}(z^{-1}) + e^{-i\pi(s-t)/2} \operatorname{Li}_{2-s-t}(z) \right\}, \quad (2.4)$$

$$C(s, t; a) = \Gamma(s + t - 1) \left\{ \frac{\Gamma(1 - t)}{\Gamma(s)} \zeta(s + t - 1, b) + \frac{\Gamma(1 - s)}{\Gamma(t)} \zeta(s + t - 1, a) \right\}. \quad (2.5)$$

All apparent singularities of the complete right side in this region are interpreted by analytic continuation. The formulas also give a meromorphic continuation in the two spectral variables.

Proof. For $\operatorname{Re} s < 0$, the classical Hurwitz Fourier series is absolutely and uniformly convergent, and its nonzero Fourier coefficients are

$$\widehat{\zeta}_s(n) = \frac{\Gamma(1-s)}{(2\pi)^{1-s}|n|^{1-s}} \exp\left(-\frac{i\pi(1-s)}{2} \operatorname{sgn} n\right), \quad n \neq 0. \quad (2.6)$$

The zero coefficient is zero. This is the exponential form of the Hurwitz Fourier identity [1, Eq. 25.11.9]. Multiplying the two absolutely convergent Fourier series and integrating leaves only frequencies n and $-n$. Translation contributes the phase $e^{-2\pi i n a}$. The sum over $n > 0$ is the first polylogarithm in (2.4), and the sum over $n < 0$ is the second.

For completeness, write $r = s + t - 1$ and insert the Hurwitz Fourier expansions of $\zeta(r, a)$ and $\zeta(r, b)$ into (2.5). Gamma reflection reduces the coefficient of $\operatorname{Li}_{1-r}(z)$ to the desired phase, using

$$\frac{\sin(\pi s)e^{-i\pi(s+t)/2} + \sin(\pi t)e^{i\pi(s+t)/2}}{\sin \pi(s+t)} = e^{-i\pi(s-t)/2}.$$

The other coefficient follows by reversing the phases. Initially avoid integer hyperplanes where this manipulation divides by zero. The result on their complement extends across removable singularities. The identity theorem and Lemma 2.1 now extend both expressions to $\operatorname{Re} s, \operatorname{Re} t < 1$. Their right sides supply the asserted meromorphic continuation. $\square$

Remark 2.3 (Translation is not ordinary reflection). The two-piece definition is not $\int_0^1 \zeta(s, x)\zeta(t, 1-x) dx$ and is not $\int_0^1 \zeta(s, x)\zeta(t, a+x) dx$ without wrapping. Replacing one by another changes the Fourier phase and the endpoint terms. Also, $a = 0$ is not a harmless substitution in the ordinary integral: the two singularities then coincide. A sufficient convergence region at $a = 0$ includes $\operatorname{Re}(s+t) < 1$ as well as $\operatorname{Re} s, \operatorname{Re} t < 1$. Section 6 handles the required continuation explicitly.

3 Endpoint subtraction and the Stieltjes generating function

The Hurwitz shift law gives the exact local decomposition

$$\gamma_m(x) = \frac{\log^m x}{x} + \gamma_m(1+x), \quad x > 0. \quad (3.1)$$

This identity determines the counterterms without a numerical fitting procedure or a choice of divergent Fourier summation.

Definition 3.1 (Separated-endpoint finite part). For integers $m, n \geq 0$ and $0 < a < 1$, define

$$\begin{aligned} I_{mn}(a) = \lim_{\epsilon \downarrow 0} \Big\{ & \int_{\epsilon}^b \gamma_m(x) \gamma_n(a+x) dx + \int_{\epsilon}^a \gamma_m(b+y) \gamma_n(y) dy \\ & + \frac{\gamma_n(a)}{m+1} \log^{m+1} \epsilon + \frac{\gamma_m(b)}{n+1} \log^{n+1} \epsilon \Big\}. \end{aligned} \quad (3.2)$$

The cutoff is smaller than $\min(a, b)$. Equivalently the two endpoints can be assigned separate cutoffs, with their respective counterterms.

Lemma 3.2 (An absolutely convergent implementation). *The limit exists and equals*

$$\begin{aligned} I_{mn}(a) &= \int_0^b \left\{ \gamma_m(x) \gamma_n(a+x) - \gamma_n(a) \frac{\log^m x}{x} \right\} dx \\ &\quad + \int_0^a \left\{ \gamma_m(b+y) \gamma_n(y) - \gamma_m(b) \frac{\log^n y}{y} \right\} dy \\ &\quad + \frac{\gamma_n(a)}{m+1} \log^{m+1} b + \frac{\gamma_m(b)}{n+1} \log^{n+1} a. \end{aligned} \quad (3.3)$$

It also satisfies $I_{mn}(a) = I_{nm}(1-a)$.

Proof. On the first interval, use (3.1). The singular remainder is $\log^m x [\gamma_n(a+x) - \gamma_n(a)]/x$, which is integrable because the bracket is $O(x)$. The other term is bounded there. The argument at the second endpoint is identical. Integrating the explicitly subtracted $\log^m x/x$ and $\log^n y/y$ terms gives (3.3). Interchange the two intervals for the symmetry. $\square$

Set

$$G(u, x) = \zeta(1+u, x) - \frac{1}{u}, \quad G(0, x) = -\psi(x), \quad g(u) = \Gamma(1-u)(2\pi)^u. \quad (3.4)$$

The removable value of G at zero is always included. Define

$$\begin{aligned} K(u, v; a) &= g(u)g(v) \{ e^{i\pi(u-v)/2} \text{Li}_{-u-v}(e^{-2\pi ia}) \\ &\quad + e^{-i\pi(u-v)/2} \text{Li}_{-u-v}(e^{2\pi ia}) \}. \end{aligned} \quad (3.5)$$

It is holomorphic near $(u, v) = (0, 0)$.

Theorem 3.3 (Holomorphic finite-part generating theorem). *Define the mixed divided difference*

$$R(u, v; a) = \frac{K(u, v; a) - K(u, 0; a) - K(0, v; a) + K(0, 0; a)}{uv}. \quad (3.6)$$

The apparent coordinate-axis singularities are removable and

$$R(u, v; a) = \sum_{m, n \geq 0} \frac{(-1)^{m+n}}{m!n!} I_{mn}(a) u^m v^n \quad (3.7)$$

in a neighborhood of the origin. In particular,

$$I_{mn}(a) = (-1)^{m+n} m!n! \left[u^{m+1} v^{n+1} \right] K(u, v; a), \quad (3.8)$$

$$I_{mn}(a) = \frac{(-1)^{m+n}}{(m+1)(n+1)} \partial_u^{m+1} \partial_v^{n+1} K(0, 0; a). \quad (3.9)$$

Proof. For $\text{Re } u, \text{Re } v < 0$, Lemma 2.1 implies

$$\int_0^1 G(u, x) G(v, x \oplus a) dx = C(1+u, 1+v; a) + \frac{1}{uv}.$$

Subtracting the endpoint poles therefore gives the meromorphic expression

$$R = C(1+u, 1+v; a) + \frac{1}{uv} + \frac{G(v, a)}{u} + \frac{G(u, b)}{v}. \quad (3.10)$$

Its holomorphy is best proved before any coefficient extraction. Put

$$T(u, c) = \frac{1 - c^{-u}}{u}, \quad T(0, c) = \log c.$$

A direct local subtraction rewrites (3.10) as

$$\begin{aligned} R = & \int_0^b \left\{ x^{-1-u} [G(v, a+x) - G(v, a)] + G(u, 1+x)G(v, a+x) \right\} dx \\ & + \int_0^a \left\{ y^{-1-v} [G(u, b+y) - G(u, b)] + G(v, 1+y)G(u, b+y) \right\} dy \\ & + G(v, a)T(u, b) + G(u, b)T(v, a). \end{aligned} \quad (3.11)$$

The differences in square brackets vanish linearly at zero. Consequently this expression is jointly holomorphic for u, v in a sufficiently small polydisc about zero; all its spectral derivatives are controlled by integrable powers times logarithms. Expanding G by (1.1) gives precisely (3.3), coefficient by coefficient, including its upper endpoint logarithms.

Theorem 2.2 and $\Gamma(-u) = \Gamma(1-u)/(-u)$ give $C(1+u, 1+v; a) = K(u, v; a)/(uv)$. Moreover the axis values are

$$K(u, 0; a) = -1 - uG(u, b), \quad K(0, v; a) = -1 - vG(v, a), \quad K(0, 0; a) = -1. \quad (3.12)$$

They follow either from (2.5) or the Gamma-ratio formula below. Substitution in (3.10) proves (3.6). Dividing a holomorphic mixed difference by uv removes exactly the axis coefficients, proving (3.8). Conversion between Taylor coefficients and derivatives gives (3.9). $\square$

Remark 3.4 (Why a finite part, rather than a product of formal series). A Fourier expansion at the Stieltjes pole is not an absolutely convergent series whose factors may be multiplied without qualification. The proof instead begins in an ordinary convergence region, continues a specified integral, and matches the continuation to local counterterms. This matching fixes the finite constant; it is not optional regularization terminology.

4 Gamma-ratio reduction and all-order linear closure

Define, with $w = u + v$,

$$A(u, v) = \frac{\Gamma(1+w)\Gamma(1-v)}{\Gamma(1+u)}, \quad B(u, v) = A(v, u). \quad (4.1)$$

Theorem 2.2 becomes the particularly useful identity

$$K(u, v; a) = -uA(u, v)\zeta(1+w, b) - vB(u, v)\zeta(1+w, a). \quad (4.2)$$

No product of two Hurwitz functions remains on the right.

Lemma 4.1 (Removal of the diagonal pole). *The function*

$$Q(u, v) = \frac{uA(u, v) + vB(u, v)}{u+v} \quad (4.3)$$

is holomorphic near zero and

$$K = -Q - uA(u, v)G(w, b) - vB(u, v)G(w, a). \quad (4.4)$$

Furthermore

$$\log A(u, v) = \sum_{k \geq 2} \frac{\zeta(k)}{k} \left\{ (-1)^k ((u+v)^k - u^k) + v^k \right\}. \quad (4.5)$$

Proof. At $v = -u$, the Gamma ratios $A(u, -u)$ and $B(u, -u)$ both equal one. The numerator in (4.3) therefore vanishes on $u + v = 0$ and is holomorphically divisible by $u + v$. Splitting off $1/w$ from each Hurwitz function gives (4.4). The expansion $\log \Gamma(1+z) = -\gamma z + \sum_{k \geq 2} (-1)^k \zeta(k) z^k / k$ in the defining ratio yields (4.5). All Euler-constant linear terms cancel exactly. $\square$

Theorem 4.2 (Bilinear closure and the missing-order rule). *For every $m, n \geq 0$, let $N = m + n + 1$. There are explicitly computable coefficients in $\mathbb{Q}[\zeta(2), \dots, \zeta(N+1)]$ such that*

$$I_{mn}(a) = e_{mn} + \sum_{j=0}^N (c_{mn,j} \gamma_j(a) + d_{mn,j} \gamma_j(b)). \quad (4.6)$$

The two highest orders satisfy

$$c_{mn,N} = \frac{1}{m+1}, \quad d_{mn,N} = \frac{1}{n+1}, \quad c_{mn,N-1} = d_{mn,N-1} = 0. \quad (4.7)$$

The symmetry is $c_{mn,j} = d_{nm,j}$ and $e_{mn} = e_{nm}$. Assigning formal weight k to $\zeta(k)$, the coefficients $c_{mn,j}, d_{mn,j}$ are homogeneous of weight $N - j$, and e_{mn} is homogeneous of weight $N + 1$. Zero coefficients obey the same convention.

Proof. Set $D = N + 1 = m + n + 2$ and $f = (-1)^{m+n} m! n!$. Let A_d be the homogeneous part of total degree d in A , and put $B_d(u, v) = A_d(v, u)$. Equation (4.4) gives the finite coefficient formulas

$$e_{mn} = -f \left[u^{m+1} v^{n+1} \right] \frac{u A_D + v B_D}{u + v}, \quad (4.8)$$

$$c_{mn,j} = -\frac{f(-1)^j}{j!} \left[u^{m+1} v^{n+1} \right] v B_{N-j}(u, v) (u + v)^j, \quad (4.9)$$

$$d_{mn,j} = -\frac{f(-1)^j}{j!} \left[u^{m+1} v^{n+1} \right] u A_{N-j}(u, v) (u + v)^j. \quad (4.10)$$

Each numerator in (4.8) is polynomially divisible by $u + v$, by homogeneous decomposition of Lemma 4.1. Every term in (4.5) has spectral degree and zeta weight equal to its index. This proves the finite coefficient algebra and weight statements.

For $j = N$, use $A_0 = B_0 = 1$. The binomial coefficient in $v(u + v)^N$ gives

$$-\frac{(-1)^{m+n+N} m! n!}{N!} \binom{N}{m+1} = \frac{1}{m+1},$$

and the other coefficient is $1/(n+1)$. For $j = N - 1$, both expressions contain A_1 or B_1 , which are zero because (4.5) has no linear term. The symmetry follows by swapping u, v and the two Stieltjes arguments. $\square$

The theorem does not treat positive zeta values as algebraically independent. Formal homogeneity describes the explicit expressions produced by the Gamma expansion. Classical relations among even zeta values may be applied afterward, or the unsimplified expressions may be retained to preserve an auditable coefficient convention.

4.1 A terminating exact coefficient algorithm

Write ℓ_k for the degree- k summand of (4.5). The homogeneous exponential satisfies

$$A_0 = 1, \quad A_1 = 0, \quad A_d = \frac{1}{d} \sum_{k=2}^d k \ell_k A_{d-k} \quad (d \geq 2). \quad (4.11)$$

To prove it, insert an auxiliary variable r into $\exp(\sum_{k \geq 2} r^k \ell_k)$, differentiate in r , and compare coefficients. For a fixed pair (m, n) , only degrees $d \leq m + n + 2$ are needed. Thus the calculation terminates by rational polynomial arithmetic, without integer-relation searches or numerical constants. The program `code/exact_coefficients.py` implements exactly (4.11) and (4.8)–(4.10).

5 Explicit identities and a convergent digamma formula

For compactness in this section only, write $a_j = \gamma_j(a)$, $b_j = \gamma_j(b)$ and $z_k = \zeta(k)$. The first identities are

$$I_{00} = a_1 + b_1 - 2z_2, \quad (5.1)$$

$$I_{10} = \frac{1}{2}a_2 + b_2 + z_2(a_0 + b_0) - z_3, \quad (5.2)$$

$$I_{11} = \frac{1}{2}(a_3 + b_3) + 2z_2(a_1 + b_1) + z_3(a_0 + b_0) - z_2^2 - z_4, \quad (5.3)$$

$$I_{20} = \frac{1}{3}a_3 + b_3 + 2z_2(a_1 + b_1) + 2z_3b_0 - z_2^2 - 3z_4, \quad (5.4)$$

$$I_{21} = \frac{1}{3}a_4 + \frac{1}{2}b_4 + 3z_2(a_2 + b_2) + 2z_3a_1 + 4z_3b_1 \\ + (2z_2^2 + 2z_4)a_0 + (z_2^2 + 3z_4)b_0 - 4z_2z_3 - 4z_5, \quad (5.5)$$

$$I_{30} = \frac{1}{4}a_4 + b_4 + 3z_2(a_2 + b_2) + 6z_3b_1 + (3z_2^2 + 3z_4)a_0 + 6z_4b_0 - 6z_5. \quad (5.6)$$

All formulas with the indices reversed follow by $a \leftrightarrow b$. For example, the lack of a_2, b_2 in I_{11} is the universal missing-order rule, not an accidental low-weight simplification.

Corollary 5.1 (Ordinary convergent digamma integrals). *For $0 < a < 1$, $b = 1 - a$,*

$$\int_0^b \left\{ \psi(x)\psi(x+a) + \frac{\psi(a)}{x} \right\} dx + \int_0^a \left\{ \psi(y)\psi(y+b) + \frac{\psi(b)}{y} \right\} dy \\ = \gamma_1(a) + \gamma_1(b) - \frac{\pi^2}{3} + \psi(a) \log b + \psi(b) \log a. \quad (5.7)$$

Both displayed integrals converge absolutely.

Proof. Insert $\gamma_0 = -\psi$ into (3.3) and use (5.1). Near zero, $\psi(x) = -1/x - \gamma + O(x)$, so each subtracted integrand is bounded. This also verifies the convergence directly, without relying on finite-part notation. $\square$

At $a = 1/2$, the distribution identity gives

$$I_{00}(1/2) = 2\gamma_1 - 4\gamma \log 2 - 2 \log^2 2 - \frac{\pi^2}{3}.$$

This follows from $\gamma_1(1/2) = \gamma_1 - 2\gamma \log 2 - \log^2 2$. The defining quadratic integral has been replaced by a short linear expression in the ordinary Stieltjes constants and elementary constants.

6 Coincident endpoints and canonical collision subtraction

A translated correlation has two separated logarithmic endpoint divergences. At zero translation they merge into an additional x^{-2} singularity. We now show that the same coefficient table still applies, but only after the merged singularity is subtracted explicitly.

For an integer $r \geq 0$, put

$$E_r(L) = \sum_{k=0}^r \frac{r!}{k!} L^k. \quad (6.1)$$

The identity $E_r - E'_r = L^r$ implies

$$\int_{\epsilon}^1 \frac{\log^r x}{x^2} dx = \frac{E_r(\log \epsilon)}{\epsilon} - r!. \quad (6.2)$$

Definition 6.1 (Coincident-endpoint finite part). Write $\gamma_j = \gamma_j(1)$ and define

$$\begin{aligned} J_{mn} = \lim_{\epsilon \downarrow 0} \left\{ \int_{\epsilon}^1 \gamma_m(x) \gamma_n(x) dx - \frac{E_{m+n}(\log \epsilon)}{\epsilon} \right. \\ \left. + \frac{\gamma_n}{m+1} \log^{m+1} \epsilon + \frac{\gamma_m}{n+1} \log^{n+1} \epsilon \right\}. \end{aligned} \quad (6.3)$$

Theorem 6.2 (Coincident closure). *The finite part (6.3) exists. With exactly the coefficients of Theorem 4.2,*

$$J_{mn} = e_{mn} + \sum_{j=0}^{m+n+1} (c_{mn,j} + d_{mn,j}) \gamma_j. \quad (6.4)$$

Equivalently it is the same mixed coefficient of

$$K_0(u, v) = -(uA(u, v) + vB(u, v))\zeta(1 + u + v). \quad (6.5)$$

In particular,

$$J_{00} = 2\gamma_1 - \frac{\pi^2}{3}. \quad (6.6)$$

Proof. By (3.1), the cutoff in Definition 6.1 converges to the absolutely convergent expression

$$\begin{aligned} J_{mn} = & -(m+n)! \\ & + \int_0^1 \left\{ \frac{\log^m x}{x} [\gamma_n(1+x) - \gamma_n] + \frac{\log^n x}{x} [\gamma_m(1+x) - \gamma_m] \right. \\ & \left. + \gamma_m(1+x) \gamma_n(1+x) \right\} dx. \end{aligned} \quad (6.7)$$

The two bracketed differences vanish linearly at zero, and (6.2) supplies the constant $-(m+n)!$.

To identify the coefficients, start where the ordinary coincident Hurwitz integral converges, for example $\operatorname{Re} s, \operatorname{Re} t < 0$. Fourier orthogonality gives

$$C_0(s, t) = 2\Gamma(1-s)\Gamma(1-t)(2\pi)^{s+t-2} \cos\left(\frac{\pi(s-t)}{2}\right) \zeta(2-s-t).$$

The Riemann functional equation, equivalently the specialization of the Fourier calculation in Theorem 2.2, gives $uvC_0(1+u, 1+v) = K_0(u, v)$. Meromorphic continuation is justified by subtracting the local terms as above.

More explicitly, the holomorphic generating expression near zero is

$$\begin{aligned} R_0(u, v) = & -\frac{1}{1+u+v} \\ & + \int_0^1 \{x^{-1-u}[G(v, 1+x) - G(v, 1)] + x^{-1-v}[G(u, 1+x) - G(u, 1)] \\ & + G(u, 1+x)G(v, 1+x)\} dx. \end{aligned}$$

The first term is the analytic continuation of $\int_0^1 x^{-2-u-v} dx$, not a discarded divergence. The remaining integrals converge locally uniformly near zero. On a starting convergence region, and hence by continuation,

$$R_0 = \frac{K_0 + 1 + uG(u, 1) + vG(v, 1)}{uv}.$$

Its Taylor coefficients are (6.7). Decomposing (6.5) as $-Q - (uA + vB)G(u + v, 1)$ now gives (6.4) by the identical finite algebra used earlier. $\square$

For the first case, an ordinary integral version is worth recording:

$$\int_0^1 \left\{ \psi(1+x)^2 - \frac{2[\psi(1+x) + \gamma]}{x} \right\} dx = 1 + 2\gamma_1 - \frac{\pi^2}{3}. \quad (6.8)$$

The apparently singular numerator in the second term vanishes at zero. There is no positivity contradiction in the negative value of J_{00} : it is a finite part of a positive divergent square, not its ordinary nonnegative integral.

6.1 An exact convergent expansion at collision

The relationship between the separated and coincident finite parts can be expressed by an equality of convergent series, not merely an asymptotic prescription.

Theorem 6.3 (Collision polynomial and convergent regular part). *For $m, n \geq 0$ define a polynomial*

$$\mathcal{P}_{mn}(L) = (-1)^{m+n+1} m!n! \left[u^{m+1} v^n \right] B(u, v) e^{-(u+v)L}. \quad (6.9)$$

It has degree $m + n + 1$, leading coefficient $1/(m + 1)$, and coefficients in positive integer zeta values. There is an exact expansion

$$I_{mn}(a) = \frac{\mathcal{P}_{mn}(\log a)}{a} + J_{mn} + \sum_{k \geq 1} h_{mn,k} a^k, \quad 0 < a < 1, \quad (6.10)$$

whose regular power series converges locally uniformly for $|a| < 1$. Its coefficients are

$$\begin{aligned} h_{mn,k} = & \frac{(-1)^{m+n+1} m!n!}{k!} \left[u^{m+1} v^{n+1} \right] \\ & (uA(u, v) + (-1)^k vB(u, v))(1 + u + v)_k \zeta(1 + u + v + k). \end{aligned} \quad (6.11)$$

For $m = n$, all odd-indexed $h_{mm,k}$ vanish. In particular,

$$\lim_{a \downarrow 0} \left\{ I_{mn}(a) - \frac{\mathcal{P}_{mn}(\log a)}{a} \right\} = J_{mn}. \quad (6.12)$$

Proof. The classical parameter Taylor series at one gives, locally uniformly in the spectral variable,

$$\zeta(1+w, 1-a) = \sum_{k \geq 0} \frac{(1+w)_k}{k!} \zeta(1+w+k) a^k,$$

while the shift law gives

$$\zeta(1+w, a) = a^{-1-w} + \sum_{k \geq 0} \frac{(1+w)_k}{k!} \zeta(1+w+k) (-a)^k.$$

These are meromorphic in w at zero, with the $k = 0$ pole explicitly visible. Substitution in (4.2) yields

$$\begin{aligned} K(u, v; a) = & -vB(u, v)a^{-1-w} \\ & - \sum_{k \geq 0} \frac{uA(u, v) + (-1)^k vB(u, v)}{k!} (1+w)_k \zeta(1+w+k) a^k. \end{aligned} \quad (6.13)$$

The $k = 0$ summand is exactly K_0 ; its apparent pole at $w = 0$ is removable. The remaining series is holomorphic for small u, v and $|a| < 1$, uniformly on compact subsets. For example, on a fixed small spectral polydisc its terms and spectral derivatives grow at most polynomially in k times powers of $\log k$, whereas $|a|^k$ decays geometrically on compact subdiscs. Coefficient extraction is therefore legitimate. The singular first term gives (6.9), the constant gives Theorem 6.2, and the other terms give (6.11).

The highest power of L comes from $B_0 = 1$ and has degree $m + n + 1$; a binomial coefficient gives its stated leading coefficient. For $m = n$ and odd k , the factor $uA - vB$ is antisymmetric under $u \leftrightarrow v$, while every other factor in (6.11) is symmetric. Its diagonal mixed coefficient is zero. Finally the regular series has zero limit at $a = 0$, proving (6.12). $\square$

The first collision polynomials are

$$\mathcal{P}_{00}(L) = L, \quad \mathcal{P}_{10}(L) = \frac{L^2}{2} + \zeta(2), \quad \mathcal{P}_{01}(L) = L^2 + \zeta(2). \quad (6.14)$$

The asymmetry of the last two is required: the factor that becomes singular at $a = 0$ has changed its index. In particular,

$$I_{00}(a) - \frac{\log a}{a} \longrightarrow 2\gamma_1 - \frac{\pi^2}{3}.$$

There is no arbitrary additional finite constant in this limit. It is fixed by the initial endpoint subtraction and agrees with the coincident prescription.

7 An ordinary log-Gamma autocorrelation identity

The same two-parameter kernel has a regular spectral specialization at zero. Unlike the Stieltjes correlations, the following integral needs no finite-part interpretation:

$$\begin{aligned} F(a) = & \int_0^b \log \Gamma(x) \log \Gamma(x+a) dx \\ & + \int_0^a \log \Gamma(b+y) \log \Gamma(y) dy, \quad 0 < a < 1. \end{aligned} \quad (7.1)$$

The endpoint singularities are logarithmic and integrable. Define $L = \log(2\pi)$, $\alpha = \gamma + L$ and $B_2(a) = a^2 - a + 1/6$.

Theorem 7.1 (Zeta and polylogarithm forms of the autocorrelation). *For $0 < a < 1$,*

$$F(a) = \frac{L^2}{4} + (1 + \zeta(2))B_2(a) - \zeta'(-1, a) - \zeta'(-1, b) - \frac{1}{2}(\zeta''(-1, a) + \zeta''(-1, b)). \quad (7.2)$$

With $z = e^{2\pi ia}$, an equivalent identity is

$$F(a) = \frac{L^2}{4} + \frac{1}{2\pi^2} \left\{ \left(\alpha^2 + \frac{\pi^2}{4} \right) \operatorname{Re} \operatorname{Li}_2(z) - 2\alpha \operatorname{Re} \partial_s^1 \operatorname{Li}_s(z) \Big|_{s=2} + \operatorname{Re} \partial_s^2 \operatorname{Li}_s(z) \Big|_{s=2} \right\}. \quad (7.3)$$

Both formulas extend continuously to $a = 0$ and $a = 1$.

Proof. Lerch's formula gives $\zeta'(0, x) = \log \Gamma(x) - L/2$ and its integral over $[0, 1]$ is zero. Differentiating (2.1) once in each spectral variable at zero is legitimate by Lemma 2.1; hence

$$F(a) = L^2/4 + \partial_s \partial_t C(0, 0; a). \quad (7.4)$$

For the first expression let $w = s + t$ and use the same Gamma ratio A as in (4.1), now in variables s, t . Equation (2.5) rewrites as

$$C(s, t; a) = \frac{sA(s, t)\zeta(w-1, b) + tB(s, t)\zeta(w-1, a)}{w(w-1)}. \quad (7.5)$$

Since $\zeta(-1, a) = \zeta(-1, b) = -B_2(a)/2$, the $w = 0$ singularity is removable. Also $A(s, t) = 1 + \zeta(2)t(s+t) + O((|s|+|t|)^3)$. Expanding the numerator through total degree three and extracting st after division gives

$$\begin{aligned} \partial_s \partial_t C(0, 0; a) &= -2(1 + \zeta(2))\zeta(-1, a) \\ &\quad - \zeta'(-1, a) - \zeta'(-1, b) - \frac{1}{2}(\zeta''(-1, a) + \zeta''(-1, b)). \end{aligned}$$

This proves (7.2).

Alternatively, differentiate the Fourier-coefficient identity (2.6) at $s = 0$. This is justified by the endpoint domination in Lemma 2.1, applied to each coefficient integral. For $n > 0$ the coefficient of the centered log-Gamma function is

$$-\frac{i}{2\pi n} \left(\alpha + \log n + \frac{i\pi}{2} \right).$$

Its product with the coefficient at $-n$ is $((\alpha + \log n)^2 + \pi^2/4)/(4\pi^2 n^2)$. The centered log-Gamma function is square-integrable, so the bilinear Parseval identity for its translation consequently gives

$$F(a) - L^2/4 = \frac{1}{2\pi^2} \sum_{n \geq 1} \frac{(\alpha + \log n)^2 + \pi^2/4}{n^2} \cos(2\pi na).$$

The series is absolutely and uniformly convergent. Expanding the square and using the spectral derivative series of Li_s at $s = 2$ gives (7.3). Uniform convergence establishes continuity on the closed interval. Equivalently, log-Gamma is square-integrable and its circular autocorrelation is continuous. $\square$

7.1 Half- and quarter-period evaluations

Write $\ell = \log 2$ and $Z_j = \zeta^{(j)}(-1)$. Differentiating the elementary Hurwitz distribution identities at -1 gives

$$F(1/2) = \frac{L^2}{4} + (1 - \ell)Z_1 + \frac{Z_2}{2} + \frac{\ell - 1 - \zeta(2)}{12} + \frac{\ell^2}{24}, \quad (7.6)$$

$$F(1/4) = \frac{L^2}{4} + \frac{Z_1}{4} + \frac{Z_2}{8} + \frac{\ell^2 - 1 - \zeta(2)}{48}. \quad (7.7)$$

For the first identity use $\zeta(s, 1/2) = (2^s - 1)\zeta(s)$. For the second use

$$\zeta(s, 1/4) + \zeta(s, 3/4) = (4^s - 2^s)\zeta(s).$$

Thus these translated products require only two Riemann-zeta jets at -1 , not an unresolved numerical relation between a large set of Gamma values.

The zero-shift limit reads

$$F(0) = \frac{L^2}{4} + \frac{1 + \zeta(2)}{6} - 2\zeta'(-1) - \zeta''(-1). \quad (7.8)$$

Differentiating the Riemann functional equation converts it to the classical quadratic log-Gamma moment recorded in the manuscript [4]:

$$\int_0^1 \log^2 \Gamma(x) dx = \frac{\gamma^2}{12} + \frac{\pi^2}{48} + \frac{\gamma L}{6} + \frac{L^2}{3} - \frac{(\gamma + L)\zeta'(2)}{\pi^2} + \frac{\zeta''(2)}{2\pi^2}.$$

This is a compatibility check, not a claim to rediscover the classical moment as a new theorem.

8 Derivatives, polygamma and exact antiderivatives

The closure theorem is useful not only because it eliminates a product. It also converts differentiation and integration of the entire family into standard operations on one Hurwitz function.

Proposition 8.1 (Parameter-derivative dictionary). *For integers $r \geq 1$, $n \geq 0$, and $a > 0$,*

$$\gamma_n^{(r)}(a) = (-1)^{r+n} n! [u^n] (1 + u)_r \zeta(r + 1 + u, a). \quad (8.1)$$

In particular,

$$\gamma_1^{(r)}(a) = (-1)^{r+1} r! (\zeta'(r + 1, a) + H_r \zeta(r + 1, a)). \quad (8.2)$$

Proof. Iterate the Hurwitz parameter derivative identity to obtain $\partial_a^r \zeta(1 + u, a) = (-1)^r (1 + u)_r \zeta(r + 1 + u, a)$. The derivative of $1/u$ in a is zero. Coefficient comparison in (1.1) gives (8.1). The first two terms of $(1 + u)_r = r!(1 + H_r u + O(u^2))$ give (8.2). $\square$

For instance,

$$\begin{aligned} \frac{d^r}{da^r} I_{00}(a) &= (-1)^{r+1} r! \{ \zeta'(r + 1, a) + H_r \zeta(r + 1, a) \\ &\quad + (-1)^r [\zeta'(r + 1, b) + H_r \zeta(r + 1, b)] \}. \end{aligned} \quad (8.3)$$

Ordinary polygamma values enter through $\psi^{(r)}(a) = (-1)^{r+1} r! \zeta(r + 1, a)$. Equations (8.1) and (4.6) give all higher derivatives of every I_{mn} without differentiating an ill-defined product at a singular endpoint.

Proposition 8.2 (A closed primitive of every correlation). *Suppose $I_{mn}(a)$ is written as in (4.6). A primitive on $0 < a < 1$ is*

$$e_{mn}a + \sum_{j=0}^{m+n+1} \frac{(-1)^{j+1}}{j+1} \left\{ c_{mn,j} \zeta^{(j+1)}(0, a) - d_{mn,j} \zeta^{(j+1)}(0, 1-a) \right\}. \quad (8.4)$$

In particular,

$$\int I_{00}(a) da = \frac{\zeta''(0, a) - \zeta''(0, 1-a)}{2} - \frac{\pi^2}{3}a + \text{constant}. \quad (8.5)$$

Proof. Expand $\partial_a \zeta(s, a) = -s\zeta(s+1, a)$ at $s = 0$. Comparing coefficients gives

$$\partial_a \zeta^{(j+1)}(0, a) = (-1)^{j+1}(j+1)\gamma_j(a).$$

This is the classical antiderivative ladder already used in the repository [4]. Apply it termwise to (8.4); the reflected argument contributes a minus sign by the chain rule. Equation (8.5) is the first specialization. $\square$

Repeated primitives can also be written with no recursively undefined functions. If $r \geq 1$ and $j \geq 0$, then

$$\partial_s^j \left\{ \frac{\zeta(s-r, a)}{\prod_{k=1}^r (k-s)} \right\} \Big|_{s=0} \quad (8.6)$$

is an r -fold primitive of $\zeta^{(j)}(0, a)$. Indeed r parameter derivatives of the numerator multiply it by $\prod_{k=1}^r (k-s)$ and restore $\zeta(s, a)$. The integration polynomial of degree at most $r-1$ can be fixed by any specified basepoint normalization. At $j = 1$, these formulas are one way to represent the negative-polygamma ladder; they do not silently change the zero-based or balanced conventions in the existing manuscript.

9 Rational grids and Dirichlet-character coordinates

The finite linear closure interacts particularly well with classical Hurwitz distribution. For integers $q \geq 2$ put

$$T_j(q) = \sum_{r=1}^{q-1} \gamma_j(r/q). \quad (9.1)$$

Proposition 9.1 (Exact rational-grid traces). *For $j \geq 0$,*

$$T_j(q) = (q-1)\gamma_j + q \sum_{k=1}^j \binom{j}{k} (-\log q)^k \gamma_{j-k} + \frac{(-1)^j q}{j+1} (\log q)^{j+1}. \quad (9.2)$$

Consequently

$$\sum_{r=1}^{q-1} I_{mn}(r/q) = (q-1)e_{mn} + \sum_{j=0}^{m+n+1} (c_{mn,j} + d_{mn,j}) T_j(q). \quad (9.3)$$

The first case is

$$\sum_{r=1}^{q-1} I_{00}(r/q) = 2(q-1)(\gamma_1 - \zeta(2)) - 2q\gamma \log q - q \log^2 q. \quad (9.4)$$

Proof. Sum the Hurwitz distribution identity and remove the $r = q$ term:

$$\sum_{r=1}^{q-1} \zeta(1+u, r/q) = (q^{1+u} - 1)\zeta(1+u).$$

The residue on both sides is $(q-1)/u$. Expanding the regular part and multiplying its u^j coefficient by $(-1)^j j!$ gives (9.2); the last term comes from the pole times the exponential q^u , and must be retained. Summing the closure formula and permuting $r \leftrightarrow q-r$ gives (9.3). $\square$

Let χ be any nonprincipal Dirichlet character modulo q , not necessarily primitive, and define $W_j(\chi) = \sum_{r=1}^{q-1} \chi(r) \gamma_j(r/q)$. Our convention uses χ , not its complex conjugate, in this sum.

Proposition 9.2 (Character resolution). *For $j \geq 0$,*

$$W_j(\chi) = (-1)^j q \sum_{k=0}^j \binom{j}{k} (\log q)^{j-k} L^{(k)}(1, \chi). \quad (9.5)$$

Moreover

$$\sum_{r=1}^{q-1} \chi(r) I_{mn}(r/q) = \sum_{j=0}^{m+n+1} (c_{mn,j} + \chi(-1) d_{mn,j}) W_j(\chi). \quad (9.6)$$

In particular the character trace of I_{mm} vanishes for every odd character.

Proof. The identity $\sum_r \chi(r) \zeta(s, r/q) = q^s L(s, \chi)$ has no pole at $s = 1$, because $\sum_r \chi(r) = 0$. Differentiating j times there gives (9.5). The constant in the closure formula sums to zero, and the reflected Stieltjes term transforms by $\chi(q-r) = \chi(-1)\chi(r)$. This proves (9.6). For $m = n$, the two coefficients are equal and their odd-character contribution cancels. $\square$

These statements reduce arithmetic traces of nonlinear-looking integrals to a finite spectral jet of a single Dirichlet L -function. They do not prove that the surviving even-character derivative coordinates are independent of other periods. No such independence assumption is used in their derivation.

10 Computational reproduction and evidence separation

The proof of Theorem 4.2 is uniform in the indices. Computations serve a different purpose: detecting coefficient, phase, normalization and implementation mistakes. The supplied programs retain this separation in their module documentation and result files.

10.1 Exact algebra

The coefficient engine generates all 45 ordered pairs (m, n) with $m + n \leq 8$. Zeta values are formal symbols during this calculation. The run recorded in `results/exact_verification.json` checks 272 exact assertions: polynomial divisibility is enforced during coefficient construction; the counted assertions verify reflection symmetry, the two highest nonzero Stieltjes coefficients, the missing next order, the leading collision coefficient, and two explicitly specified base identities. The result is PASS.

The full data in `results/coefficients.json` store the constant, the coefficients of $\gamma_j(a)$ and $\gamma_j(1-a)$, and the collision polynomial for every pair. The companion `results/identities.txt` is a human-readable rendering. No transcendental equality is inferred from rational rank computations, and the symbolic checks are not a formalization of the convergence proofs.

10.2 Independent numerical diagnostics

With Python 3.13.5, mpmath 1.3.0 and 45 decimal working digits, `code/verify_numeric.py` passed 19 checks. They compare ordinary or explicitly subtracted quadrature with the closed formulas, compare polylogarithm and Hurwitz kernels at noninteger real and complex orders, check rational traces, and differentiate a closed antiderivative. The largest recorded absolute residual in this run is approximately 1.06×10^{-44} .

A separate program, `code/verify_stieltjes_pairs.py`, evaluates the regularized products I_{10}, I_{11}, I_{20} at $a = 1/2$ by quadrature. It evaluates the necessary Stieltjes functions through a 140-term parameter Taylor expansion at one, using Hurwitz spectral derivatives for the coefficients, rather than evaluating the closed correlation formula inside its own integrand. The recorded comparisons are

Quantity	Quadrature value (rounded)	Absolute residual
$I_{10}(1/2)$	6.71092887447221586072	3.98×10^{-41}
$I_{11}(1/2)$	-8.64046177754241055451	1.61×10^{-40}
$I_{20}(1/2)$	-11.02758300286848051739	2.34×10^{-40}

The Taylor truncation and quadrature errors are not enclosed by directed interval arithmetic. The table is therefore numerical evidence, not certified error bounds. The combined numerical count is 22, separate from the 272 exact symbolic checks.

A negative control omits the term $-2\zeta(2)$ from I_{00} and verifies that the resulting discrepancy is $2\zeta(2)$. This tests the particular finite constant that is easy to lose when endpoint subtraction is replaced by an informal cancellation of divergent Fourier terms.

10.3 A reproducible order-differentiation canary

At integer spectral orders, differentiating a polylogarithm numerical routine with default finite-difference settings can lose substantially more digits than direct function evaluation suggests. The exact identity

$$\operatorname{Re} \operatorname{Li}_s(i) = 2^{-s}(2^{1-s} - 1)\zeta(s) \quad (10.1)$$

for real s near 2 supplies a simple test. It follows by grouping the absolutely convergent series by even indices, and then continues in s .

In the installed mpmath 1.3.0, at 40 working digits and with the exact argument `mp.mpc(0,1)`, applying the default `mp.diff` to the left and right sides of (10.1) gives zero recorded discrepancy for the values, but absolute discrepancies

$$2.33945455584 \times 10^{-12} \quad \text{and} \quad 2.44970539683 \times 10^{-16}$$

for the first and second spectral derivatives, respectively. The minimal reproducer and full observed values are supplied in `code/polylog_derivative_canary.py` and its JSON output.

This is a version- and method-specific observation, not a counterexample to (10.1), nor a claim that every polylogarithm implementation has the same behavior. The validating log-Gamma tests therefore use exact cyclotomic reduction before spectral differentiation at i and -1 . The raw polylogarithm kernel is also checked separately at noninteger spectral parameters. Future numerical implementations may remove the canary discrepancy; the reproducer records it rather than requiring it to persist.

11 Audit, normalization safeguards and integration scope

Finite part versus ordinary integral. The new formulas must be integrated with Definition 3.1, not with a bare integral sign. Counterterms include both endpoint coefficients and the correct powers of logarithms. At collision, the extra $\epsilon^{-1}E_{m+n}(\log \epsilon)$ term is indispensable. For users who prefer ordinary integrals, (3.3), (5.7) and (6.7) eliminate finite-part notation entirely.

Analytic continuation and diagonal cancellation. Gamma poles in (2.5) and the apparent pole at $u + v = 0$ in (4.2) cancel only in the combined expression. Expanding or evaluating the individual terms at a pole is invalid. The regular expression (4.4) or exact homogeneous coefficients should be used near the intersection of the axes. The limit $(1 - c^{-u})/u \rightarrow \log c$ also fixes an easily reversed sign in subtraction code.

Differentiating singular products. The parameter derivatives in Section 8 differentiate a proved closed expression. We do not assert that differentiation commutes with an unmodified finite-part product integral. Moving the break point, differentiating endpoint values, and differentiating counterterms can all contribute. A separate commutation theorem would be required for that claim.

Repository status and existing corrections. The inspected integration material already distinguishes classical identities from experimental searches, and its denominator-five log-Gamma correction retains the required Dirichlet derivative [4]. We do not replace that term by an unsupported Clausen-only reduction. The current exact S_4 proof must retain its proved status [5]. Neither the S_6 problem nor period independence has been settled by the present bilinear calculus.

Computational finding, not an invented source erratum. The observed polylogarithm order-derivative accuracy loss is a real reproducible finding in the delivery environment. It motivates a regression canary for future repository computations. We have not established that a particular existing repository result relied on the failing default differentiation path, so no existing theorem is retracted on that basis.

Review boundaries. The read of `main` at the end of the review returned commit `e1294e75708b84986bfdaf103892ac9ac78a0580`. Individual selected file blob identifiers are recorded separately in `notes/PROVENANCE.json`; the earlier reads were not all pinned to that commit. Incoming ZIPs were inventoried but not unpacked or audited. The package proposes a new report and a namespaced integration section, without modifying any live repository file. This respects the incoming archive distinction documented by the repository [7].

12 Further research questions

12.1 A trilinear analogue and its genuine new coordinates

For three distinct translations, define a product finite part by subtracting the local singular terms at all three endpoints. Does its spectral generating function reduce to finitely many Tornheim-type or colored double-zeta jets, and what is a useful exact coefficient normal form? Unlike the bilinear Fourier calculation, a trilinear integral retains a nontrivial additive relation among three frequencies. A linear closure theorem analogous to Theorem 4.2 should not be presumed without proving those remaining relations.

12.2 Unequal integer dilations

Replace the translated factor by a periodic function of $qx + a$, or allow different positive integer dilations in the two factors. Fourier orthogonality imposes a divisibility condition instead of equality of frequencies. Determine a finite distribution formula for the resulting finite parts, including the endpoints introduced by the dilation. This could connect the Gamma-ratio coefficient algebra directly to the manuscript's rational-grid modules.

12.3 Parameter differentiation with moving counterterms

Prove an explicit commutation law between differentiation in a and the finite-part operation. The desired theorem should list every endpoint term and yield independent integral identities involving higher polygamma and Stieltjes derivatives, rather than merely restating the closed parameter derivative in Section 8.

12.4 Higher integral spectral resonances

The expansions here are centered at spectral pairs $(1, 1)$ and $(0, 0)$. At other integer pairs, Gamma factors and Hurwitz poles can meet in more complicated ways. Classify the corresponding removable combinations and determine which products reduce to finite single-function jets. The first useful examples involve a Stieltjes factor paired with a polygamma or an explicitly normalized negative-polygamma factor.

12.5 Compatibility of successive renormalized convolutions

Theorem 6.3 proves compatibility of one collision with a specified finite part. It does not establish associativity of successive renormalized products. Determine whether an associative convolution operation can be defined on a suitable space of periodic distributions, and identify any finite local terms needed to recover the present normalization.

12.6 Arithmetic traces beyond the linear closure

Character traces of the bilinear family reduce to ordinary Dirichlet L -jets. Which additional coordinates survive for trilinear products, unequal dilations, or nonrational shifts? Any claim of irreducibility should be stated in an explicit formal law quotient unless numerical period independence has actually been proved.

12.7 Priority comparison, formalization and exact certificates

Compare the all-order coefficient theorem systematically with the wider Hurwitz-transform and finite-part-integral literature. A result already implicit in a classical source should receive appropriate attribution, while the explicit counterterm and coefficient presentation can still serve as a useful repository contribution. Formalization can then separate the local analytic lemmas, Fourier orthogonality, meromorphic continuation and finite homogeneous algebra. The last layer already has replayable exact data, but the present package is not a proof-assistant certificate.

The existing colored-polylogarithm problems, including the next mixed Gaussian reduction after S_4 , remain distinct. They may benefit from spectral generating functions, but no implication from this closure theorem to their resolution is asserted.

A Reproduction commands and artifact map

From the package root, the mathematical checks are run by

```
python code/exact_coefficients.py --max-total 8
python code/verify_numeric.py --dps 45
python code/verify_stieltjes_pairs.py
python code/polylog_derivative_canary.py
```

The tested dependencies are SymPy 1.14.0 and mpmath 1.3.0. The scripts write JSON result records and return a nonzero exit code when a required identity check fails; the version-specific canary records its observation without assuming that future software must reproduce the same loss. For the PDF, use

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The Makefile exposes these as `make check` and `make pdf`. The integration fragment defines its own auxiliary macro names and uses the label prefix `corrclosure:`. It is a proposed mathematical section, not an automatic patch against a moving repository snapshot. The article itself is self-contained and requires no external figure or bibliography files.

B Coefficient data conventions

Each row of `results/coefficients.json` contains the integer indices `m`, `n`, a constant polynomial, an array `a` of coefficients of $\gamma_j(a)$, an array `b` of coefficients of $\gamma_j(1-a)$, and a polynomial `collision`. Within these strings, `z2`, `z3`, and so on denote Riemann zeta values; `L` denotes $\log a$ only in the collision polynomial. This differs from the local notation $L = \log(2\pi)$ used in Section 7. The distinction is explicit in the JSON schema and must be preserved when importing the data.

The symbolic table intentionally retains expressions such as $\zeta(2)^2$ and $\zeta(4)$ separately. One may subsequently apply $\zeta(2)^2 = \frac{5}{2}\zeta(4)$, but doing so changes the presentation rather than the theorem. The coefficient generator does not use any unproved independence conjecture to certify the rows.

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